

Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools

A Comparative Review of Empirical Benchmarks, Mitigation Strategies, and Open Research Questions

Abstract

Large language models (LLMs) and the agentic search and “deep research” systems built on top of them are increasingly used to locate, summarize, and cite scholarly and journalistic sources, yet a substantial body of empirical work shows that these systems routinely fabricate the citations they produce. This paper synthesizes benchmark evidence on citation fabrication rates across general-purpose chatbots (GPT-3.5, GPT-4, GPT-4o, Gemini/Bard, Claude, Grok, DeepSeek), retrieval-augmented and agentic search products (Perplexity, Bing/Copilot, ChatGPT Search, Elicit, SciSpace), and domain-specific systems in law and medicine. Reported fabrication rates vary widely — from under 15% to over 90% — depending on model generation, prompt specificity, domain familiarity, and whether the system is closed-book or retrieval-grounded. We review the taxonomies used to define citation hallucination, the methodologies (human verification, DOI/PubMed cross-checking, natural-language-inference-based claim verification, and dynamic benchmark generation) used to measure it, and the current landscape of detection and mitigation tools, including ALCE, HalluLens, GhostCite/CiteVerifier, and the Reference Hallucination Score. We then discuss methodological limitations that hinder cross-study comparison, including inconsistent operational definitions, non-representative sampling of domains and prompts, and rapid model turnover that dates benchmark results within months. We close by identifying research gaps — standardized cross-model benchmarking protocols, longitudinal tracking, and evaluation of multi-step agentic search pipelines rather than single-turn generation — and propose directions for future work aimed at making citation reliability a measurable, comparable, and improvable property of deployed AI systems.

Keywords

large language models; citation fabrication; hallucination; retrieval-augmented generation; agentic search; benchmark evaluation; reference verification; AI reliability

1. Introduction

The rapid deployment of large language models (LLMs) as research assistants, literature-review aids, and “AI search engines” has created a new class of failure mode with direct consequences for scholarly and public trust: the generation of citations to sources that do not exist, or that exist but do not support the claim attached to them. This behavior is commonly grouped under the broader umbrella of *hallucination* — model output that is inconsistent with the model’s training data, the retrieved evidence, or verifiable facts about the world (Huang et al., 2025; Bang et al., 2025). Within this broader phenomenon, *citation fabrication* (also called “reference hallucination” or “ghost citation” generation) is a distinct and unusually

consequential subtype because a fabricated citation carries the outward form of verifiability — an author list, a title, a venue, a DOI or PMID — while offering none of its substance. A citation that looks legitimate is, from a user’s perspective, harder to distrust than an unsupported prose claim (Walters & Wilder, 2023; Alkaissi & McFarlane, 2023).

The practical stakes of this failure mode are no longer hypothetical. In *Mata v. Avianca, Inc.*, attorneys submitted a federal court brief containing six fabricated case citations generated by ChatGPT, resulting in judicial sanctions and international press coverage (Mata v. Avianca, Inc., 2023). Chief Justice John Roberts subsequently cited hallucination risk in his annual report on the federal judiciary. Parallel incidents have been documented in mental-health research (Linardon et al., 2025), orthopedic and maxillofacial surgery literature reviews (Rajalekshmi & Agrawal, 2026; Chelli et al., 2024), economics scholarship (citations in the Journal of Economic Literature domain), and peer review at major machine-learning venues, where fabricated references have appeared in both submitted and accepted papers.

A second and more recent development compounds the problem: the shift from closed-book chatbots to *agentic search* systems — tools such as Perplexity, ChatGPT Search, Microsoft Copilot, and “deep research” agents that autonomously issue web queries, retrieve documents, and synthesize answers with inline citations. These systems were widely assumed to be substantially more reliable than closed-book generation because citations are nominally grounded in retrieved text. Benchmark evidence reviewed in this paper shows that assumption to be only partially correct: retrieval reduces but does not eliminate fabrication, and in some evaluations agentic search tools misattribute or invent sources at rates comparable to, or exceeding, those of the underlying base models (Jaźwińska & Chandrasekar, 2025; Magesh et al., 2024).

This paper has three aims. First, we synthesize the empirical literature that has measured citation fabrication rates across general-purpose LLMs, retrieval-augmented systems, and agentic search tools, presenting a comparative view of reported rates and the conditions under which they were measured. Second, we review the methodological landscape — how researchers define, operationalize, and benchmark citation fabrication, from manual DOI verification to automated natural-language-inference (NLI) pipelines and dynamically generated benchmarks. Third, we identify the principal research gaps that currently prevent rigorous, comparable, longitudinal measurement of this problem, and we propose directions for future benchmark design.

2. Background and Related Work

2.1 Hallucination taxonomies and where citation fabrication fits

Huang et al. (2025) provide a widely cited taxonomy distinguishing *factuality hallucinations* (contradicting verifiable real-world facts) from *faithfulness hallucinations* (deviating from user instructions, context, or logical consistency). Bang et al. (2025), in introducing the HalluLens benchmark, propose a complementary and more operational split between *intrinsic* hallucination (output inconsistent with the given input context, as in summarization) and *extrinsic* hallucination (output that can be neither supported nor refuted by the model’s training data — for example, an invented paper). Citation fabrication is

predominantly an extrinsic phenomenon in closed-book generation, but becomes an intrinsic-faithfulness problem in retrieval-augmented settings, where a model may cite a retrieved document that does not, in fact, support the attached claim (Gao et al., 2023). This distinction matters because closed-book fabrication and retrieval-grounded misattribution require different detection strategies and are not always separated in benchmark reporting, which is one source of the wide variance in reported rates discussed in Section 3.

2.2 Early documentation of citation fabrication

The earliest systematic evidence emerged shortly after ChatGPT’s public release. Alkaissi and McFarlane (2023) documented that all five references ChatGPT generated for a medical summary had nonexistent titles, with PubMed identifiers pointing to unrelated papers; when re-prompted, the model produced different fabricated titles attached to the same mismatched identifiers. Athaluri et al. (2023) analyzed 178 references generated for research proposals and found that a majority lacked a valid DOI or could not be located via search. Walters and Wilder (2023), in the most cited systematic study of this early period, generated 84 short literature reviews across 42 multidisciplinary topics using GPT-3.5 and GPT-4 and manually verified all 636 resulting citations, finding that 55% of GPT-3.5 citations and 18% of GPT-4 citations were entirely fabricated, with substantive errors (wrong volume, issue, or page numbers) present in a further 43% and 24%, respectively, of the non-fabricated references. Gravel, D’Amours-Gravel, and Osmanliu (2023) reported similar patterns for medical questions, coining the phrase “learning to fake it” to describe the model’s tendency to respond with confident, well-formatted, but invented citations rather than declining to answer.

2.3 Domain-specific replications

Subsequent work extended this basic finding across disciplines and model generations. In psychiatry, ChatGPT and Bard were found to fabricate the majority of citations offered during a simulated literature search, with Bard initially providing 0% accurate plain-text citations before improving somewhat when restricted to hyperlinked PubMed references. Chelli et al. (2024) replicated eleven human-conducted systematic reviews on shoulder rotator-cuff pathology using ChatGPT and Bard and found hallucination rates ranging from 28.6% to 91.3% of returned references depending on model and prompt specificity. Linardon et al. (2025) systematically varied topic familiarity and prompt specificity within a single domain (mental health) and found that GPT-4o fabricated 19.9% of citations overall, but that the rate rose from roughly 6% for a well-studied condition (major depressive disorder) to 28–29% for less-studied conditions (binge-eating disorder, body dysmorphic disorder), with a further 64% of fabricated citations that included a DOI pointing to a real but entirely unrelated paper — a pattern that makes fabrication substantially harder for a reader to detect than an obviously invented reference. In law, Dahl et al. (2024) analyzed responses to roughly 200,000 verifiable legal queries per model and found hallucination rates of 58% (GPT-4) to 88% (Llama 2), rising to at least 75% for questions about a case’s core holding — with all tested models performing close to chance on the harder task of judging the precedential relationship between two cases.

3. Comparative Citation Fabrication Rates: LLMs vs. Agentic Search Tools

Table 1 consolidates fabrication-rate figures reported across the studies reviewed for this paper. The figures are not directly comparable in a strict statistical sense — studies differ in domain, prompt design, sample size, verification methodology, and the precise operational definition of “fabricated” — but the aggregate picture is informative: no system evaluated to date, including retrieval-grounded commercial products marketed specifically on the strength of their sourcing, achieves a fabrication rate close to zero.

Benchmark / Study	Systems Evaluated	Reported Fabrication Rate	Domain / Notes
Walters & Wilder (2023)	GPT-3.5, GPT-4	55% (GPT-3.5); 18% (GPT-4)	Multidisciplinary literature reviews (42 topics, 636 citations)
Athaluri et al. (2023)	GPT-3 (ChatGPT)	~54% lacked a valid DOI or match	Research-proposal references (178 citations)
Agrawal et al. (2024)	GPT-3, GPT-3.5, GPT-4	~70% (GPT-3) to <50% (GPT-4)	Reference-title hallucination; self-consistency probing
Linardon et al. (2025)	GPT-4o	19.9% overall; up to 29% (low-visibility topics)	Mental-health literature reviews; topic-familiarity effect
Chelli et al. (2024)	ChatGPT (GPT-3.5/4), Bard	28.6%–91.3% of returned references	Systematic-review replication (shoulder pathology)
Aljamaan et al. (2024) – RHS	ChatGPT, Bing, Bard, Perplexity, Elicit, SciSpace	Highest: ChatGPT/Bing; lowest: Elicit/SciSpace	Medical prompts; introduces Reference Hallucination Score
Cabezas-Clavijo & Sidorenko-Bautista (2025)	ChatGPT, Claude, Copilot, DeepSeek, Gemini, Grok, Le Chat, Perplexity	39.8% erroneous/fabricated; only 26.5% fully correct	400 academic references across 5 disciplines
Rajalekshmi & Agrawal (2026)	ChatGPT, Gemini, Perplexity	Perplexity highest RHS (6.51); ChatGPT lowest (1.81)	Rotator-cuff clinical literature (3,150 references)
Dahl et al. (2024)	GPT-3.5, GPT-4, PaLM 2, Llama 2	58–88% of legal queries hallucinated	Case citation, holding, and authorship recall
Magesh et al. (2024)	Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law AI	17–33% hallucination despite RAG grounding	Commercial legal-RAG products
Jaźwińska & Chandrasekar (2025) – Tow Center	ChatGPT Search, Perplexity (Pro),	Incorrect on 60%+ of queries (37%	Live news-citation retrieval (1,600 queries, 20 publishers)

Benchmark / Study	Systems Evaluated	Reported Fabrication Rate	Domain / Notes
	Gemini, DeepSeek, Copilot, Grok-2/3	Perplexity – 94% (Grok-3)	
GhostCite / CiteVerifier (2026)	13 state-of-the-art LLMs	14.23%–94.93% across models	40 research domains; 2.2M citation cross-validation

Table 1. Reported citation fabrication / hallucination rates across selected benchmark studies. Figures are drawn from the cited primary sources; definitions of “fabrication” vary by study (see Section 5).

3.1 The generation-over-generation improvement, and its limits

A consistent pattern across the general-purpose LLM literature is that fabrication rates fall from one model generation to the next: Walters and Wilder (2023) report a drop from 55% (GPT-3.5) to 18% (GPT-4), and Agrawal, Suzgun, Mackey, and Kalai (2024) report a comparable decline from over 70% (GPT-3) to under 50% (GPT-4) using a self-consistency probing method. This trend is frequently cited as evidence that the problem is being “engineered away.” The more recent, larger-scale GhostCite/CiteVerifier study complicates this narrative: benchmarking thirteen contemporary state-of-the-art LLMs across forty research domains, it finds fabrication rates spanning 14.23% to 94.93%, indicating that domain and prompt conditions can dominate model-generation effects, and that even 2025–2026-era frontier models remain capable of near-total fabrication under adverse conditions. Model-family comparisons in a global-governance benchmarking study similarly found large, statistically significant differences in fabrication propensity between contemporaneous open-weight models (roughly 11% to 30% for one class of fabrication measure across four models of comparable release date), underscoring that fabrication rate is not simply a function of scale or recency but of training data, alignment procedure, and citation-specific tuning.

3.2 Agentic and retrieval-augmented search tools

Retrieval-augmented generation (RAG) and agentic search were designed in part to solve the citation-fabrication problem by grounding output in retrieved documents rather than parametric memory alone. The evidence for how well this works is mixed and system-dependent. Aljamaan et al. (2024), introducing the Reference Hallucination Score (RHS), tested six chatbots on identical medical prompts and found that the retrieval-oriented academic tools Elicit and SciSpace produced the lowest hallucination scores, while ChatGPT and Bing produced the highest, with Perplexity intermediate — suggesting that retrieval grounding helps only when paired with conservative, domain-restricted retrieval rather than open web search. Rajalekshmi and Agrawal (2026) instead found Perplexity to have the *highest* hallucination score of the three tools tested (ChatGPT, Gemini, Perplexity) on rotator-cuff literature, exceeding both general-purpose chatbots — a result the authors attribute to Perplexity’s more aggressive live web retrieval surfacing lower-quality or tangential sources that are then cited with unwarranted confidence.

The most methodologically rigorous evaluation of agentic news search comes from the Tow Center for Digital Journalism’s Columbia Journalism Review study (Jaźwińska & Chandrasekar, 2025), which issued 1,600 identical queries (200 per tool across eight tools: ChatGPT Search, Perplexity, Perplexity Pro, Gemini,

DeepSeek Search, Copilot, Grok-2 Search, and Grok-3 Search) built from verbatim excerpts of 200 news articles guaranteed to rank in the top three organic Google results. Collectively, the tools answered more than 60% of queries incorrectly; Grok-3 was wrong on 94% of queries, while the best performer, Perplexity, was still wrong 37% of the time. Critically, the study found that (a) licensing partnerships between AI companies and publishers did not reliably predict citation accuracy for that publisher’s content; (b) premium, paid tiers of chatbots gave *more* confidently incorrect answers than their free counterparts in several cases; and (c) tools were systematically bad at declining to answer when they lacked reliable information, preferring to fabricate a plausible citation over expressing uncertainty — ChatGPT, for instance, signaled low confidence in only fifteen of two hundred responses despite 134 incorrect attributions. A companion Tow Center study focused specifically on ChatGPT Search found that when a publisher’s crawler access was blocked, the tool frequently fabricated source material rather than disclosing the retrieval failure, with this behavior present in more than a third of reviewed responses (Jaźwińska & Chandrasekar, 2024).

3.3 Domain-specialized retrieval: legal research tools

Legal research provides the clearest natural experiment on whether commercial RAG grounding solves fabrication, because vendors have made explicit reliability claims that researchers were able to test directly. Magesh, Surani, Dahl, Suzgun, Manning, and Ho (2024) conducted the first pre-registered evaluation of Lexis+ AI, Westlaw AI-Assisted Research, and Ask Practical Law AI — commercial products marketed using language such as “hallucination-free” linked citations — against 202 hand-scored legal queries. Hallucination rates of 17% to 33% were observed even in these RAG-grounded, purpose-built legal products, substantially lower than the 58–88% observed for general-purpose closed-book models by the same research group (Dahl et al., 2024), but far from the vendors’ claims. The study further distinguished *correctness* from *groundedness* failures: a response can cite a real, retrieved case while still misdescribing or “misgrounding” what that case actually held, a failure mode invisible to citation-existence checks alone and requiring claim-level verification against the cited source’s text.

4. Current Approaches and Developments in Measurement and Mitigation

4.1 Benchmark and evaluation frameworks

- ALCE (Automatic LLMs’ Citation Evaluation) (Gao et al., 2023) was the first reproducible benchmark for citation-grounded long-form question answering, combining three datasets (ASQA, QAMPARI, ELI5) with automatic metrics for fluency, correctness, and citation quality; even the best systems evaluated provided complete citation support for only about 50% of statements on the ELI5 dataset.
- HalluLens (Bang et al., 2025) addresses benchmark saturation and data leakage by dynamically generating test instances (e.g., plausible but nonexistent entity names) rather than relying on fixed, potentially memorized question sets, and formalizes the extrinsic/intrinsic taxonomy referenced in Section 2.

- The Reference Hallucination Score (RHS) (Aljamaan et al., 2024) operationalizes fabrication along six bibliographic dimensions (existence, author accuracy, title accuracy, journal accuracy, year accuracy, and topical relevance) plus a composite severity score, enabling finer-grained comparison than a binary fabricated/real label; it has since been adopted or adapted in orthopedic (Rajalekshmi & Agrawal, 2026) and other clinical-literature replications.
- GhostCite / CiteVerifier (2026) is, to date, the largest cross-domain audit, combining a 13-model, 40-domain generation benchmark with a retrospective analysis of 2.2 million citations from 56,381 papers at top-tier AI/ML and security venues, finding that roughly 1% of published papers already contain invalid or fabricated references — evidence that the problem has begun to leak from LLM outputs into the permanent scholarly record.
- AuthorityBench (2026) isolates the causal effect of citation presence itself, using a balanced 2×2 factorial design (true/false claim × real/fabricated citation) across seven models and roughly 220,000 prompts; it finds that attaching any citation — real or fabricated — increases downstream hallucination relative to an uncited claim, and that the effect is largest, not smallest, when a fabricated citation is attached to an otherwise true claim, with larger models showing no more resistance than smaller ones.
- Domain-specific benchmarks such as LegalCiteBench (2026) isolate closed-book citation recovery from retrieval-augmented performance; across 21 LLMs tested on roughly 24,000 instances built from real U.S. judicial opinions, even the strongest models scored below 7 out of 100 on exact citation retrieval, with misleading-answer rates exceeding 94% for 20 of 21 models on retrieval-heavy sub-tasks.

4.2 Detection and mitigation techniques

Beyond benchmarking, a parallel line of work targets automatic detection and correction. Agrawal et al. (2024) proposed indirect-query self-consistency checks — re-querying the model about a citation’s existence through paraphrased, indirect prompts — reporting a 20–30% reduction in false-discovery rate relative to direct verification prompts, and observing that models are considerably better at recalling the true author list of a *real* citation than at producing a consistent author list for a fabricated one, implying that some latent signal of non-existence is present in the model even when it does not surface spontaneously. Gao et al. (2023) propose a Generate-then-Refine pipeline that appends and prunes citations post hoc using retrieval and natural-language-inference (NLI) entailment checks without altering the underlying response text. Multi-agent verification pipelines (e.g., the CiteAudit approach) decompose the problem into claim extraction, evidence retrieval, passage-level matching, and calibrated judgment, explicitly targeting the *groundedness* failure mode identified by Magesh et al. (2024) rather than mere citation existence. At the product level, RAG deployment itself functions as a partial mitigation — as reflected in the consistently lower rates for retrieval-grounded academic tools (Elicit, SciSpace) relative to open-domain chatbots in the RHS study — though Sections 3.2 and 3.3 show this benefit is inconsistent across tools and can be reversed when retrieval surfaces low-quality sources.

5. Research Challenges and Limitations

5.1 Inconsistent operational definitions

The single largest obstacle to comparing fabrication rates across the studies in Table 1 is definitional inconsistency. Some studies count a citation as “fabricated” only if no matching publication can be found under any close variant of the given metadata; others count any citation with two or more incorrect bibliographic fields (author, title, year) as fabricated, which blurs the boundary between fabrication and mere citation error (Chelli et al., 2024). Studies also differ on whether a real citation attached to an unsupported or misdescribed claim — a groundedness failure — is scored as equivalent to a wholly invented reference. Because these are meaningfully different failure modes with different causes and different mitigations, aggregating them into a single “hallucination rate” obscures more than it reveals.

5.2 Non-representative and rapidly stale sampling

Most benchmark studies rely on a small, hand-curated set of prompts or topics (often fewer than 50) applied to a handful of models tested at a single point in time. Given documented sensitivity to topic familiarity (Linardon et al., 2025), prompt specificity, and domain (GhostCite’s 14.23%–94.93% range across just 40 domains), single-snapshot studies with narrow domain coverage risk both over- and understating a system’s general reliability. Compounding this, the pace of model release makes any fixed-model benchmark stale within months: a study evaluating GPT-4 in early 2024 says little about GPT-4o or GPT-5-class models released a year later, yet vendor and press claims frequently cite older benchmark figures as though they were current.

5.3 Verification methodology and scalability

Manual verification against library databases, DOI resolvers, and PubMed remains the gold standard for accuracy but does not scale to the millions of citations now being produced daily, and it is itself imperfect: fabricated citations that reuse a real journal name with an invented article title, or attach a real DOI to an unrelated paper (a pattern Linardon et al., 2025, found in 64% of fabricated citations with DOIs), are specifically designed — in the sense of being a natural byproduct of next-token prediction over plausible-sounding text — to survive superficial automated checks such as DOI-resolves-to-something tests. Automated NLI-based and multi-agent verification pipelines (Section 4.2) improve scalability but introduce their own error rates and have not yet been benchmarked against a common, independently audited gold-standard corpus across the full range of study types.

5.4 Confounding of retrieval quality with model reliability

Because agentic search tools bundle a retriever, a ranking or filtering step, and a generator into a single user-facing product, published “fabrication rates” for tools like Perplexity or Copilot conflate at least three distinct sources of error: retrieval failure (relevant source not found), ranking failure (irrelevant source surfaced and cited), and generation failure (source retrieved correctly but claim misrepresented or citation invented outright). The Tow Center and RHS studies document behaviorally distinct patterns — Copilot’s unusually high non-answer rate despite full crawler access, Perplexity’s comparatively better

performance on some evaluations and worse on others — that are consistent with this conflation but that existing benchmark designs are not built to decompose.

5.5 Publisher, licensing, and access confounds specific to agentic search

Agentic web-search tools introduce a confound absent from closed-book LLM evaluation: publisher-side access controls (robots.txt, crawler blocking) and commercial licensing agreements directly shape what content a tool can retrieve, independent of model quality. Jaźwińska and Chandrasekar (2025) found that licensing deals did not reliably predict citation accuracy and that blocked crawler access sometimes correlated with *increased* fabrication as tools attempted to answer anyway rather than declining, meaning fabrication-rate comparisons across tools partly reflect each vendor’s distinct crawling and licensing footprint rather than a pure measure of generation quality.

6. Research Gaps and Future Directions

- **Standardized, shared benchmark protocols:** The field lacks a single, widely adopted protocol analogous to TruthfulQA or MMLU for citation fabrication specifically. A shared benchmark with an explicit, granular taxonomy (existence, bibliographic accuracy, topical relevance, and claim-level groundedness, following the RHS and AuthorityBench models) evaluated identically across closed-book and agentic-search systems would enable genuine cross-study comparison.
- **Longitudinal and versioned tracking:** Given the demonstrated month-to-month volatility of fabrication rates across model releases, the field would benefit from a continuously updated, dynamically regenerated leaderboard (in the spirit of HalluLens’s leakage-resistant test generation) that tracks fabrication rates by model version over time, analogous to existing hallucination leaderboards for general factuality.
- **Decomposing agentic pipelines:** Future benchmark designs should isolate retrieval failure, ranking/filtering failure, and generation-time fabrication or misgrounding as separate, independently measured stages, rather than reporting a single end-to-end error rate that conflates them — particularly important as “deep research” agents add multi-step planning, tool use, and iterative query reformulation on top of single-shot RAG.
- **Domain and language coverage:** Existing benchmarks skew heavily toward English-language biomedical, legal, and computer-science content. Systematic coverage of underrepresented domains (humanities, Global South publication venues, non-English scholarship) is needed, particularly given evidence that DOI and citation hallucination rates differ sharply by field and region (e.g., markedly higher humanities DOI-hallucination rates relative to natural sciences, and higher fabrication rates for Global South relative to Global North content in at least one governance-focused benchmark).
- **Causal and mechanistic understanding:** AuthorityBench’s finding that citation presence itself — independent of its truth value — increases hallucination on adjacent claims suggests citation fabrication is entangled with broader authority-bias and sycophancy dynamics rather than being a purely retrieval- or memory-bound problem; mechanistic work connecting this to model internals (e.g., hallucination-associated activation patterns) is at an early stage and warranted.

- User-facing calibration and abstention: Multiple studies (Tow Center; Dahl et al., 2024) find that models rarely hedge or decline to answer even when producing confidently wrong citations. Research on training and evaluating appropriate abstention specifically for citation tasks — rather than general factual QA — remains underdeveloped relative to its practical importance for downstream trust.
- Downstream contamination monitoring: With GhostCite/CiteVerifier documenting fabricated citations already present in published, peer-reviewed venues, there is a clear need for standardized post-publication citation-integrity auditing tools integrated into journal and conference submission pipelines, rather than relying solely on pre-generation model improvements.

7. Conclusion

Citation fabrication is not a transient artifact of early, under-trained language models; it is a persistent property of both closed-book LLMs and the retrieval-augmented, agentic search tools built to overcome the closed-book setting’s limitations. Reported fabrication rates across the studies reviewed here span roughly 15% to over 90%, shaped less by whether a system nominally performs web retrieval than by domain familiarity, prompt specificity, retrieval quality, and the presence or absence of calibrated abstention behavior. Retrieval grounding measurably reduces — but does not eliminate — fabrication, and can in some configurations introduce new failure modes (low-quality source surfacing, misgrounding of accurately retrieved sources) that are invisible to simple citation-existence checks. Commercial claims of “hallucination-free” citation, tested directly in the legal domain, have not withstood empirical scrutiny. Progress will depend less on a single model-scaling breakthrough than on the maturation of standardized, dynamically updated, and mechanistically decomposed benchmarks capable of tracking this fast-moving problem across model generations, domains, and the increasingly agentic architectures through which people now access AI-mediated information.

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